Appendix VII. Debt Sustainability Analysis

Palau's public debt is expected to remain sustainable despite the AsDB loans of about US$53 million which started to be disbursed in FY2016. Under the baseline scenario, the public debt-to-GDP ratio is projected to decline to 25.6 percent of GDP by FY2023, mainly driven by the scheduled debt repayment for major project loans starting from FY2018. GDP growth shocks and associated fiscal costs arising from high volatility in the tourism sector or natural disasters are the major risks to debt sustainability.

1. The public debt-to-GDP ratio started to increase starting from FY2016 and it is projected to continue to increase through FY2018, before declining gradually afterwards, due to the disbursements from the recent AsDB loans.[1] The disbursements of AsDB loans for the Koror-Airai sanitization and submarine-cable projects started in FY2016 and are planned to be completed by FY2019. Public debt in the analysis includes the central government's external debt (19.9 percent of GDP in FY2017), the government's domestic accounts payable or domestic arrears (4.2 percent of GDP in FY2017), and the SOEs' debt (9.8 percent of GDP in FY2017), which is fully guaranteed by the government and is a contingent liability.

2. An important element in the analysis is that Palau does not have access to private foreign capital markets. All of Palau's external debt is from international organizations for specific projects, and thus there is no direct linkage between the fiscal balance and debt issuance in the private debt market. The baseline scenario assumes that the central government and SOEs do not take new external loans during FY2018–23, and thus the debt stock declines in line with the amortization of past loans. The path of public debt could be less favorable if the government or the SOEs take new debts (e.g., Koror-Airai sanitization project-phase II, US$ 25million) in the coming years.

3. Under the baseline scenario, the public debt-to-GDP ratio is projected to decline to 25.6 percent of GDP by FY2023 from 34.5 percent in FY2018. The decline of the public debt-to-GDP ratio will be driven by the scheduled debt repayment for major project loans during this period. As the primary surpluses and gains from the growth-interest rate differential in FY2020–23 are larger than those required to pay for the scheduled debt service, the government uses the surplus to fully repay its domestic arrears and to build government deposits.

4. The alternative scenarios such as the constant primary balance scenario highlight Palau's stable fiscal position in that these scenarios do not lead to large deviations from public debt and gross financing needs under the baseline scenario. However, the GDP growth shock scenario and the historical scenario highlight the sensitiveness of the public debt-to-GDP ratio and financing needs to a growth shock. The GDP growth shock scenario of a decline in real GDP growth by 5.2 percentage points (corresponding to one standard deviation of GDP growth during past 10 years) during FY2019–20 would lead the public debt-to-GDP ratio to jump to 46.4 percent of GDP in FY2020 and remains around 40 percent by FY2023.

[1] Public debt in the DSA is larger than that in Table 4 because it covers both central government and SOEs which hold a large share of public debt in recent years.

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5. This result highlights that GDP growth shocks and associated fiscal costs—which may result from the high volatility of the tourism sector and large natural disasters—are the major risks to the public-sector debt sustainability.[2] Also, the potential increase in government expenditure from several sources (the contingent liability shock) would be another source of risk for debt sustainability.[3]

[2] Lee et al. (2018) shows that extreme natural disasters may have significant impacts on GDP growth and fiscal balance in Pacific island countries with varying degrees: 0.4~5.6 percentage point decline in GDP growth and 0.9~4.1 percentage point decline in fiscal balance-to-GDP ratio (IMF Working Paper, 18/108).

[3] The scenario considers the potential increase in government expenditure arising from the failure of repayment on the on-lent funds by the Palau Public Utilities Corporation (PPUC) and Palau National Communications Corporation (PNCC) as well as the potential government support to the Civil Service Pension Fund (CSPF)'s obligations (assumed as 2 percent of GDP per year in total). However, the potential liabilities from public-private partnership project (e.g., PPP for airport renovation and expansion) were not included due to lack of information to identify the size of contingent liabilities.

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Figure 1. Palau: Public Sector Debt Sustainability Analysis (DSA) - Baseline Scenario

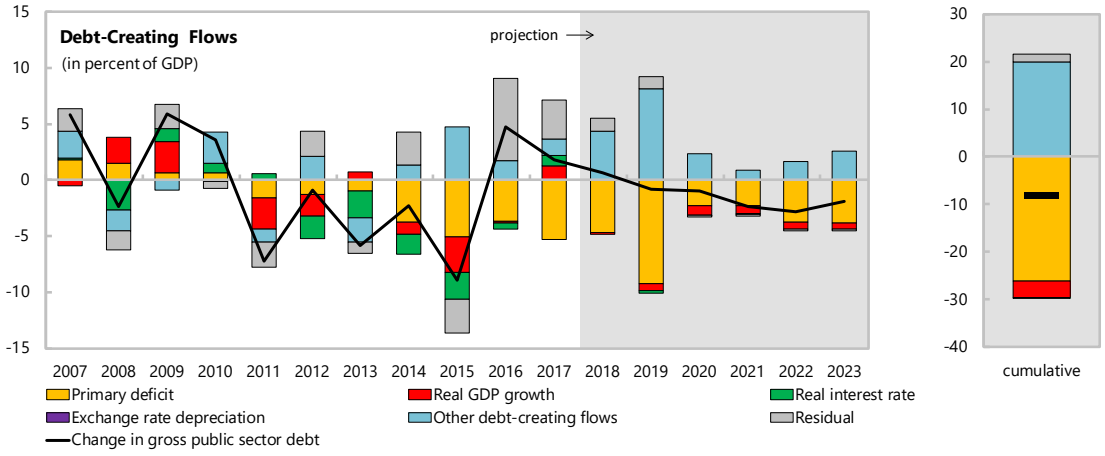
(In percent of GDP, unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

	Actual			Projections						As of September 06, 2018		
	2007-2015 [2/]	2016	2017	2018	2019	2020	2021	2022	2023			
Nominal gross public debt	42.3	32.0	33.8	34.5	33.6	32.7	30.3	27.5	25.6	Sovereign Spreads		
Of which: guarantees	20.2	10.6	9.8	8.5	8.2	7.8	7.5	7.2	6.9	EMBIG (bp) 3/		n.a.
Public gross financing needs	0.1	-2.8	-4.0	-3.9	-6.7	1.1	2.1	-0.1	-1.4	5Y CDS (bp)		n.a.
Real GDP growth (in percent)	1.4	0.5	-3.7	0.4	2.0	2.5	2.2	2.0	2.0	Ratings	Foreign	Local
Inflation (GDP deflator, in percent)	3.8	3.2	-1.0	1.9	2.3	2.0	2.0	2.0	2.0	Moody's	n.a.	n.a.
Nominal GDP growth (in percent)	5.3	3.7	-4.7	2.3	4.4	4.5	4.2	4.0	4.0	S&Ps	n.a.	n.a.
Effective interest rate (in percent) [4/]	1.2	1.2	2.0	2.0	1.8	1.9	1.9	2.1	2.2	Fitch	n.a.	n.a.

Contribution to Changes in Public Debt

	Actual			Projections							
	2007-2015	2016	2017	2018	2019	2020	2021	2022	2023	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	-1.4	4.8	1.8	0.6	-0.8	-1.0	-2.3	-2.8	-1.9	-8.2	
Identified debt-creating flows	-1.5	-2.6	-1.7	-0.5	-1.9	-0.8	-2.2	-2.7	-1.8	-9.8	
Primary deficit	-0.9	-3.7	-5.3	-4.7	-9.2	-2.3	-2.3	-3.8	-3.8	-26.2	2.1
Primary (noninterest) revenue and grants	43.6	41.0	39.8	40.0	49.3	41.2	40.7	40.5	40.4	252.2	
Primary (noninterest) expenditure	42.7	37.3	34.5	35.3	40.0	38.9	38.4	36.8	36.6	226.0	
Automatic debt dynamics [5/]	-1.4	-0.7	2.2	-0.1	-0.8	-0.8	-0.7	-0.6	-0.5	-3.6	
Interest rate/growth differential [6/]	-1.4	-0.7	2.2	-0.1	-0.8	-0.8	-0.7	-0.6	-0.5	-3.6	
Of which: real interest rate	-1.0	-0.5	1.0	0.0	-0.2	0.0	0.0	0.0	0.1	-0.2	
Of which: real GDP growth	-0.4	-0.1	1.2	-0.1	-0.7	-0.8	-0.7	-0.6	-0.5	-3.4	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.8	1.7	1.4	4.3	8.2	2.4	0.9	1.7	2.5	19.9	
Change in government domestic currency a	0.8	1.7	1.4	4.3	8.2	2.4	0.9	1.7	2.5	19.9	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Other debt flows (e.g., ESM and Euroarea lo	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	0.1	7.4	3.5	1.1	1.1	-0.1	-0.1	-0.1	-0.2	1.6	



Source: IMF staff.

1/ Public sector is defined as central government and includes public guarantees, defined as SOE Debt.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes changes in the stock of guarantees, asset changes, and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

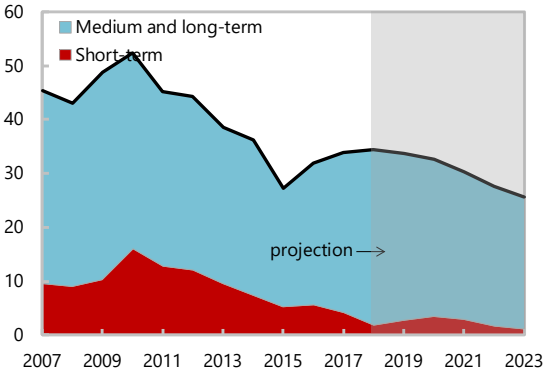
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Figure 2. Palau: Public DSA—Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

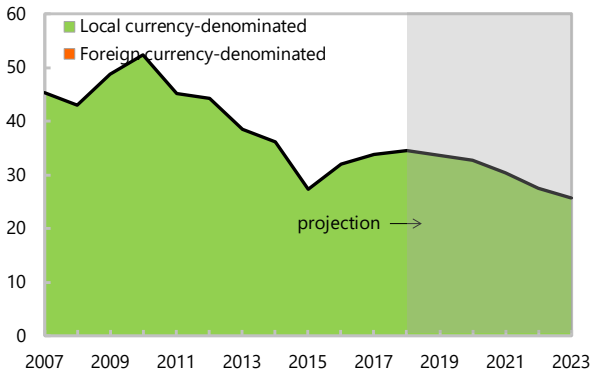
By Maturity

(in percent of GDP)

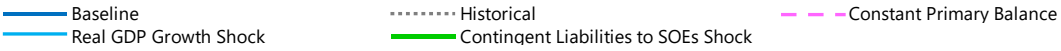


By Currency

(in percent of GDP)

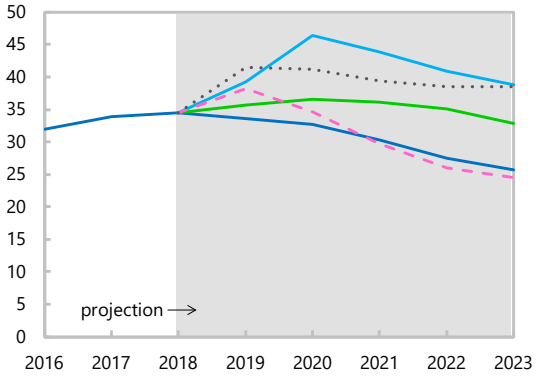


Alternative Scenarios



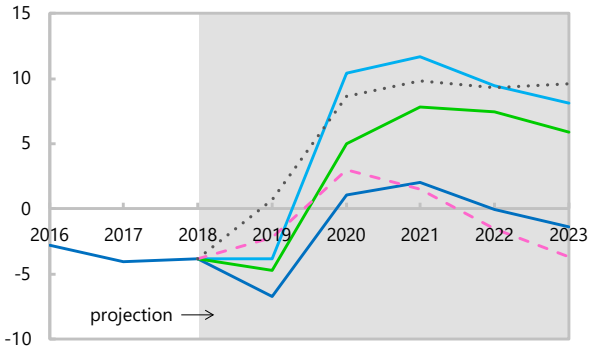
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2018	2019	2020	2021	2022	2023
Real GDP growth	0.4	2.0	2.5	2.2	2.0	2.0
Inflation	1.9	2.3	2.0	2.0	2.0	2.0
Primary Balance	4.7	9.2	2.3	2.3	3.8	3.8
Effective interest rate	2.0	1.8	1.9	1.9	2.1	2.2
Constant Primary Balance Scenario						
Real GDP growth	0.4	2.0	2.5	2.2	2.0	2.0
Inflation	1.9	2.3	2.0	2.0	2.0	2.0
Primary Balance	4.7	4.7	4.7	4.7	4.7	4.7
Effective interest rate	2.0	1.8	1.6	1.8	2.1	2.4
Contingent Liability shock						
Real GDP growth	0.4	2.0	2.5	2.2	2.0	2.0
Inflation	1.9	2.3	2.0	2.0	2.0	2.0
Primary Balance	4.7	7.2	2.3	2.3	3.8	3.8
Effective interest rate	2.0	1.8	1.8	1.7	1.6	1.6

Historical Scenario	2018	2019	2020	2021	2022	2023
Real GDP growth	0.4	0.8	0.8	0.8	0.8	0.8
Inflation	1.9	2.3	2.0	2.0	2.0	2.0
Primary Balance	4.7	1.9	1.9	1.9	1.9	1.9
Effective interest rate	2.0	1.8	0.9	0.8	0.9	0.8
Real GDP Growth Shock						
Real GDP growth	0.4	-3.2	-2.7	2.2	2.0	2.0
Inflation	1.9	1.0	0.7	2.0	2.0	2.0
Primary Balance	4.7	6.5	-3.1	2.3	3.8	3.8
Effective interest rate	2.0	1.8	1.8	1.9	1.5	1.6

Source: IMF staff.

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